

The Orthogonal-Circle Ruled Surface: Immersion Theory, a Topological Dipole, and Projection Singularities

Akari Hayami (Jian-Yu Huang)

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Abstract

We study the ruled surface $\mathcal{S}$ in $\mathbb{R}^3$ defined by two mutually orthogonal unit circles subject to the pairing constraint $\cos t + \cos \phi = 1$. All results are independently derived from first principles. Three separate calculations reveal a single governing polynomial $\mathcal{P}(c) = c^2 - 3c + 1$ (where $c = \cos t$): it appears as the numerator of the scalar triple product certifying non-developability (Theorem 2.1), as the equation governing the isolated immersion singularities (Theorem 2.3), and as the annihilator of the complex observation field $F = N + iM$ (Theorem 4.2). The unique root $c_0 = (3 - \sqrt{5})/2 = \varphi^{-2}$ —involving the inverse square of the golden ratio—is a canonical geometric parameter of $\mathcal{S}$. The field F carries a topological dipole: the Jacobian determinant at its two zeros evaluates exactly to $\pm 5s_0(1 - c_0)^2/Q_0$, yielding indices $\text{Ind}(P_{\pm}) = \mp 1$ and total charge zero. The projection Jacobian J_F vanishes on a fold locus governed by the triple-angle polynomial $c^3 - 3c + 1$, whose root $c_* = 2\cos(4\pi/9) \approx 0.347$ bifurcates the locus into disconnected branches. The phase skeleton $\mathcal{L}_{\text{skel}}$ and fold locus $\mathcal{L}_{\text{fold}}$ intersect at a root of a quintic equation, confirming a deep coupling between observation degeneracy and phase-flow separation (Theorem 4.4). Near the fold, the Whitney normal form $(\xi, \eta) \mapsto (\xi, \eta^2)$ induces frequency doubling and pseudo-recurrence in the projected signal from purely geometric rigid-body motion, without any physical oscillator.

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1 Definition and Parametrization

Definition 1.1 (Generating circles). *Let*

$$C_1(t) = (\cos t, \sin t, 0)^\top, \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

and

$$C_2(\phi) = (1 + \cos \phi, 0, \sin \phi)^\top, \quad \phi \in [0, \pi],$$

be two unit circles in mutually perpendicular planes, with C_1 centred at the origin and C_2 centred at $(1, 0, 0)$. The orthogonal-circle ruled surface $\mathcal{S}$ is defined by the single-parameter pairing constraint

$$\cos t + \cos \phi = 1. \tag{1}$$

Remark 1.2. *The classical Oloid satisfies the ruling condition $\cos u = -\cos t/(1 + \cos t)$ [1], inequivalent to (1). Hence $\mathcal{S}$ is non-developable (Theorem 2.1).*

Throughout we abbreviate $c := \cos t$, $s := \sin t$. Constraint (1) gives $\cos \phi = 1 - c$ and, taking $\sin \phi \geq 0$:

$$Q(t) := \sqrt{2c - c^2} = \sqrt{c(2 - c)} \geq 0,$$

which is real for $c \in [0, 1]$, confirming the domain $t \in [-\pi/2, \pi/2]$.

Definition 1.3 (Parametrization). *With ruling parameter $u \in [0, 1]$, the surface is*

$$S(t, u) = (1 - u) C_1(t) + u C_2(\phi(t)) = \begin{pmatrix} c + 2u(1 - c) \\ (1 - u)s \\ uQ(t) \end{pmatrix}. \tag{2}$$

For each fixed t , the map $u \mapsto S(t, u)$ is affine; hence $\mathcal{S}$ is a ruled surface.

2 Algebra and Intrinsic Geometry

Differentiating $Q^2 = c(2 - c)$ with respect to t :

$$Q_t = \frac{s(c - 1)}{Q}. \tag{3}$$

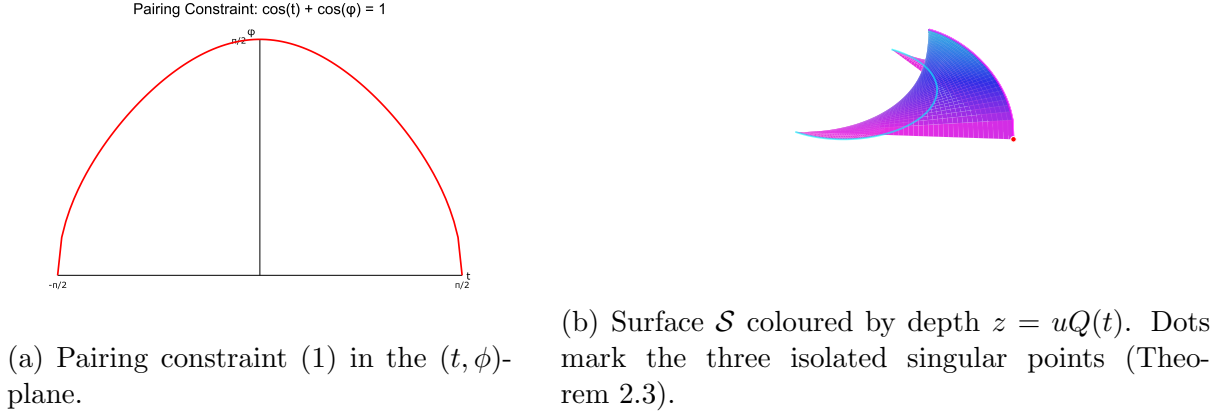


Figure 1: The orthogonal-circle ruled surface and its defining constraint.

The tangent vectors are

$$S_u = (2(1 - c), -s, Q)^\top, \quad (4)$$

$$S_t = \left((2u - 1)s, (1 - u)c, \frac{us(c-1)}{Q} \right)^\top, \quad (5)$$

with mixed partial $S_{tu} = d' := (2s, -c, s(c - 1)/Q)^\top$.

2.1 Non-developability

Theorem 2.1 (Non-developability).

$$\det(S_t, S_u, S_{tu}) = -\frac{(c^2 - 3c + 1)s}{Q(t)}. \quad (6)$$

This is not identically zero; hence $\mathcal{S}$ is non-developable with non-zero Gaussian curvature almost everywhere.

Proof. Write $S = \gamma(t) + u d(t)$ with $\gamma = C_1$ and $d = S_u$. Then $S_t = \gamma' + u d'$, $S_{tu} = d'$, so by multilinearity

$$\det(S_t, S_u, S_{tu}) = \det(\gamma', d, d') + u \underbrace{\det(d', d, d')}_{=0} = \det(\gamma', d, d').$$

Expanding the 3×3 determinant along its third row (which contains the zero entry of $\gamma' = (-s, c, 0)^\top$):

$$\det(\gamma', d, d') = scQ + \frac{s(c-1)}{Q}(s^2 - 2c(1-c)).$$

Collecting over denominator Q and substituting $Q^2 = c(2-c)$, $s^2 = (1-c)(1+c)$:

$$= \frac{s}{Q}[c^2(2-c) - (1-c)^3].$$

Expanding: $(2c^2 - c^3) - (1 - 3c + 3c^2 - c^3) = -c^2 + 3c - 1 = -(c^2 - 3c + 1)$, yielding (6). The zero set $\{s = 0\} \cup \{c^2 - 3c + 1 = 0\}$ has measure zero. $\square$

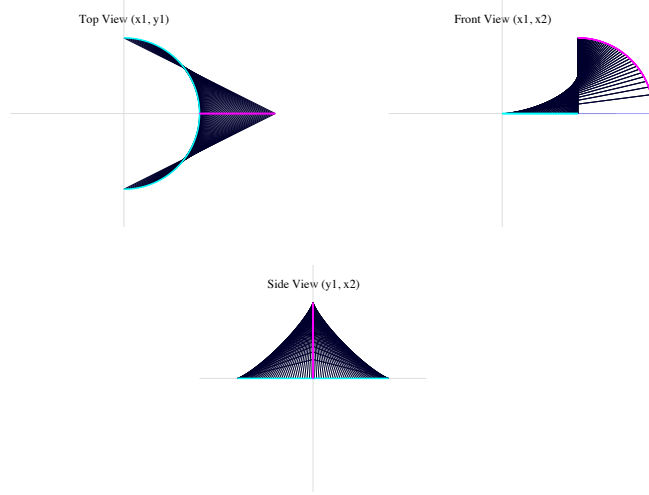


Figure 2: Orthographic three-views of $\mathcal{S}$: top (x, y) -plane (horseshoe plan), front (x, z) -plane (central spine and cusp), side (y, z) -plane (saddle twist).

2.2 First Fundamental Form

Proposition 2.2 (First fundamental form). *The metric coefficients of $\mathcal{S}$ are*

$$G(t) = \langle S_u, S_u \rangle = 2 \cos^2 t - 6 \cos t + 5, \quad (7)$$

$$F(t, u) = \langle S_t, S_u \rangle = s[(3 - 2c)u - (2 - c)], \quad (8)$$

$$E(t, u) = \langle S_t, S_t \rangle = 1 + (-4s^2 - 2c^2)u + \left(4s^2 + c^2 + \frac{s^2(1-c)^2}{c(2-c)}\right)u^2. \quad (9)$$

For $c \in [0, 1]$, $G(t) \in [1, 5]$, so $G > 0$ globally: the surface has no ruling singularities and undergoes non-uniform intrinsic stretching [4, Ch. 3].

Proof. (G) $G = 4(1 - c)^2 + s^2 + Q^2 = 4 - 8c + 4c^2 + (1 - c^2) + c(2 - c) = 2c^2 - 6c + 5$. Completing the square: $G = 2(c - \frac{3}{2})^2 + \frac{1}{2} \geq \frac{1}{2} > 0$, with $G(0) = 5$ and $G(\pm\pi/2) = 1$.

(F) Direct expansion using $Q_t Q = s(c - 1)$ gives the stated formula.

(E) Expand $\|S_t\|^2$ as a quadratic in u ; at $u = 0$: $S_t = (-s, c, 0)$ gives $E = 1$. $\square$

2.3 Regularity Classification

Theorem 2.3 (Regularity classification). *Let $V = S_t \times S_u$. The surface $\mathcal{S}$ is a regular immersion everywhere except at exactly three isolated singular points where $V = 0$:*

1. the polar singularity at $(t, u) = (0, 1)$;

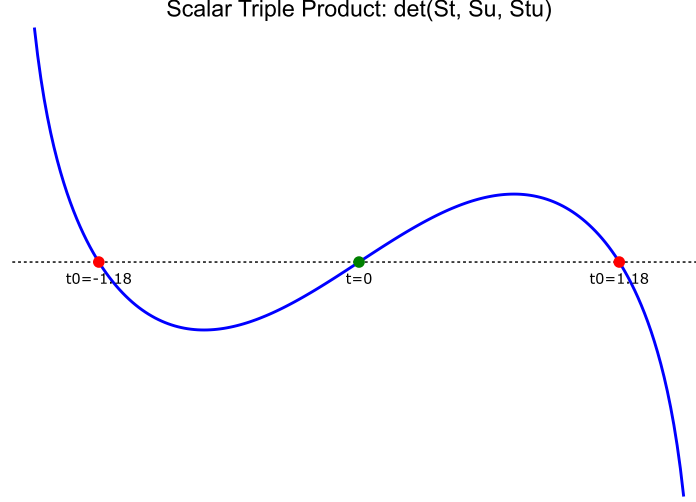


Figure 3: Scalar triple product (6) as a function of t . Sign changes at $\pm t_0$ where $c_0 = \varphi^{-2} \approx 0.382$, the root of the core invariant $c^2 - 3c + 1$.

2. the two lateral singularities at

$$(t, u) = (\pm t_0, u_0), \quad t_0 = \arccos\left(\frac{3-\sqrt{5}}{2}\right), \quad u_0 = \frac{\sqrt{5}-1}{4}.$$

The lateral parameter $c_0 = (3 - \sqrt{5})/2 = \varphi^{-2}$ is the inverse square of the golden ratio.

Proof. Direct computation of the three components of $V = S_t \times S_u$:

$$V_z = (1 - c)(1 - c - 2u), \quad (10)$$

$$V_y = -\frac{s}{Q} [2u - c(2 - c)], \quad (11)$$

$$V_x = \frac{(1 - u)c^2(2 - c) - u(1 - c)^2(1 + c)}{Q}. \quad (12)$$

Derivation of V_z : $-(2u - 1)(1 - c)(1 + c) - 2(1 - u)(1 - c)c$. Factor $(1 - c)$; the bracket simplifies to $1 - c - 2u$.

Derivation of V_y : $\frac{-2us(1-c)^2 - (2u-1)sQ^2}{Q}$. Substituting $Q^2 = c(2 - c)$, the uc^2 terms cancel leaving $c(2 - c) - 2u$.

Derivation of V_x : $\frac{(1-u)cQ^2 + us^2(c-1)}{Q}$. Use $s^2(c - 1) = -(1 - c)^2(1 + c)$ and $Q^2 = c(2 - c)$.

Case 1: $c = 1$ ($t = 0$, $s = 0$). $V_z = V_y = 0$ automatically. $V_x = (1 - u)/Q_0 = 0$ iff $u = 1$. This gives the polar singularity.

Case 2: $c \neq 1$, $s \neq 0$. $V_z = 0 \Rightarrow u = (1 - c)/2$. $V_y = 0 \Rightarrow u = c(2 - c)/2$. Equating:

$$\frac{1-c}{2} = \frac{c(2-c)}{2} \implies c^2 - 3c + 1 = 0.$$

The unique root in $(0, 1)$ is $c_0 = (3 - \sqrt{5})/2$, giving $u_0 = (\sqrt{5} - 1)/4$.

Closed-loop check for V_x : With $1 - u_0 = (1 + c_0)/2$, $V_x = \frac{1+c_0}{2Q_0} [c_0^2(2 - c_0) - (1 - c_0)^3]$. The bracket equals $-c_0^2 + 3c_0 - 1 = -(c_0^2 - 3c_0 + 1) = 0$, so $V_x = 0$ holds automatically. $\square \quad \square$

3 The Core Algebraic Invariant

Remark 3.1 (Unifying polynomial). *The polynomial $\mathcal{P}(c) = c^2 - 3c + 1$ with unique root $c_0 = \varphi^{-2} \approx 0.382$ in $(0, 1)$ has appeared in three independent derivations: as the numerator of (6) (Theorem 2.1), from the system $V_y = V_z = 0$ (Theorem 2.3), and—as shown in §4—from $N = M = 0$ (Theorem 4.2). A complementary polynomial $c^3 - 3c + 1$ (expressible via a triple-angle identity) separately controls the fold locus (Theorem 4.3).*

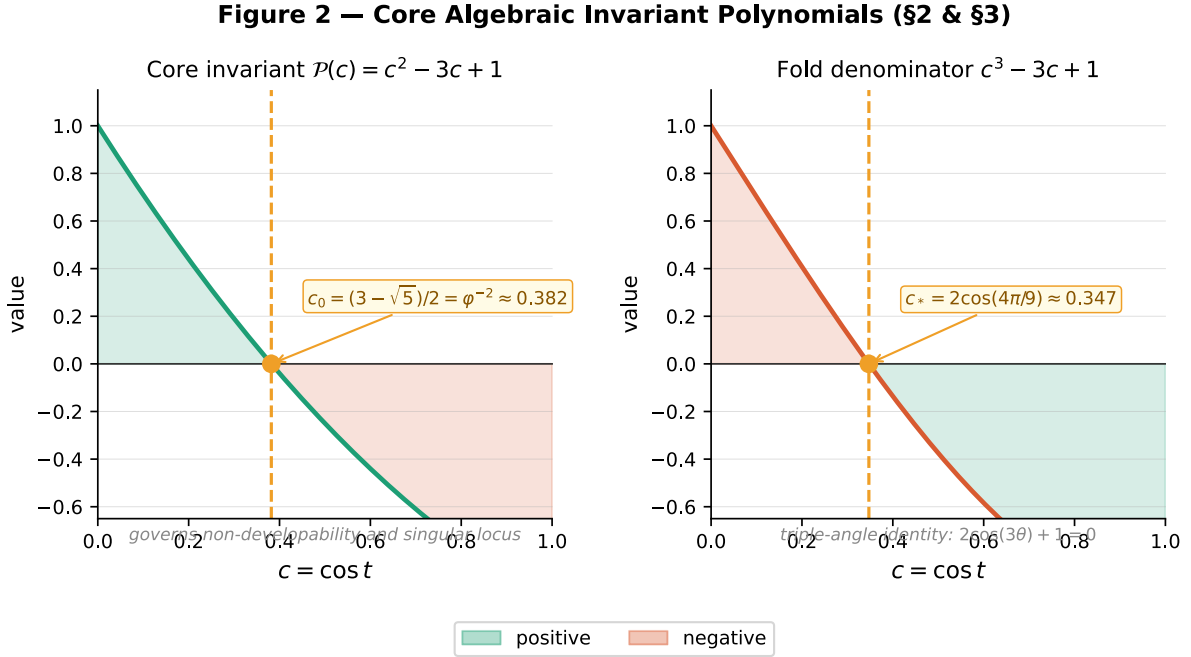


Figure 4: Left: core invariant $\mathcal{P}(c) = c^2 - 3c + 1$ (root $c_0 \approx 0.382$) governing non-developability, immersion singularities, and the topological dipole. Right: fold denominator $c^3 - 3c + 1$ (root $c_* \approx 0.347$) producing the triple-angle asymptote of the fold locus. Sign regions shaded.

4 Complex Observation Field and Projection Singularities

4.1 The Complex Observation Field

Embed $S \in \mathbb{R}^3$ trivially into $\mathbb{R}^4$. Under a rotation $R_y(\theta) \in SO(3)$ about the y -axis, the coordinates projected onto the (x_1, y_1) -plane are

$$X_\theta = x_1 \cos \theta + x_2 \sin \theta, \quad Y_\theta = y_1,$$

where $x_1 = c + 2u(1 - c)$, $y_1 = (1 - u)s$, $x_2 = uQ$. The Jacobian of $(t, u) \mapsto (X_\theta, Y_\theta)$ decomposes as

$$J_\theta = N(t, u) \cos \theta - M(t, u) \sin \theta, \quad (13)$$

where, remarkably, N and M coincide with cross-product components from §2:

$$N(t, u) = V_z = (1 - c)(1 - c - 2u), \quad M(t, u) = V_x = \frac{(1 - u)c^2(2 - c) - u(1 - c)^2(1 + c)}{Q}. \quad (14)$$

Define the *complex observation field*

$$F(t, u) = N(t, u) + i M(t, u). \quad (15)$$

Since $F = 0$ forces $V_y = 0$ via (11), we have $F = 0 \Leftrightarrow V = 0$: the zeros of F coincide exactly with the immersion singularities.

4.2 Phase Skeleton

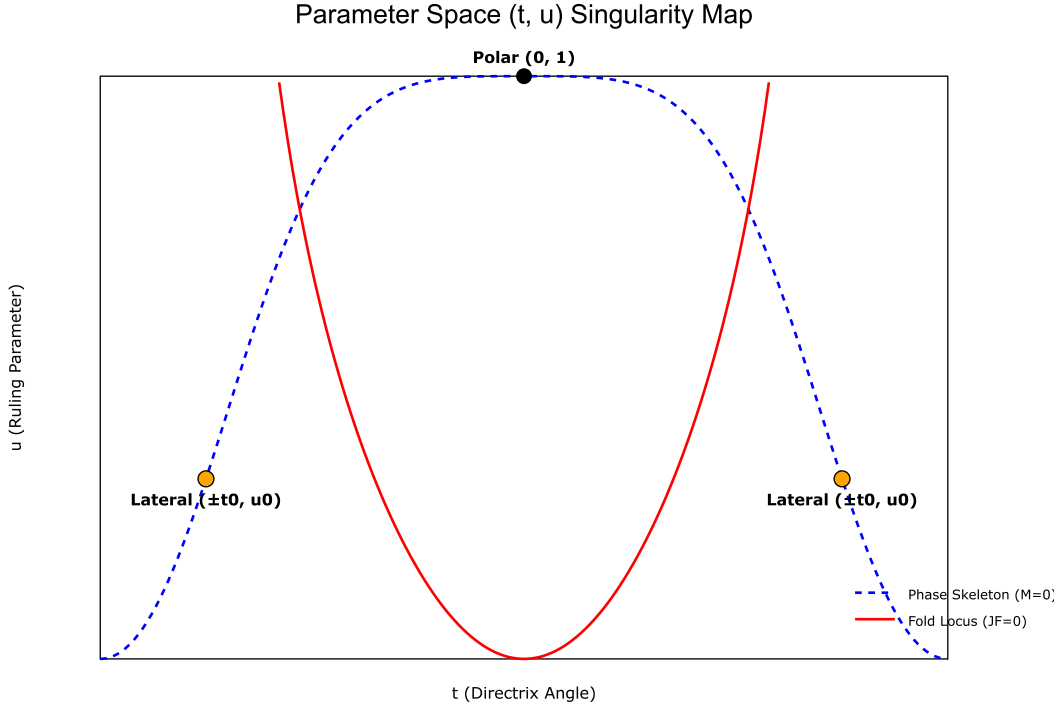


Figure 5: Overview of the parameter domain (t, u) : phase skeleton $\mathcal{L}_{\text{skel}}$ (teal) and fold locus $\mathcal{L}_{\text{fold}}$ (coral) as curves in the angular coordinate t . Orange dots: the topological dipole zeros $P_{\pm}$ at $(\pm t_0, u_0)$. The absolute singularity (Theorem 4.4) lies at their intersection.

Proposition 4.1 (Phase skeleton). *The locus $\mathcal{L}_{\text{skel}} := \{M = 0\}$ is the rational curve*

$$u_{\text{skel}}(c) = \frac{c^2(2 - c)}{1 - c + c^2}. \quad (16)$$

The denominator $1 - c + c^2 = (c - \frac{1}{2})^2 + \frac{3}{4} \geq \frac{3}{4} > 0$ for all real c , so $\mathcal{L}_{\text{skel}}$ is smooth and spans the full parameter domain, partitioning it into phase sectors $\{\arg F = 0\}$ and $\{\arg F = \pi\}$.

Proof. Set $M = 0$ and solve for u : $c^2(2-c) = u[c^2(2-c) + (1-c)^2(1+c)]$. Expanding the denominator: $(2c^2 - c^3) + (1 - 2c + c^2)(1+c) = 1 - c + c^2$ (the c^3 and two c^2 cross-terms cancel), yielding (16). $\square$

4.3 Topological Dipole

Theorem 4.2 (Topological dipole). *The complex field $F = N + iM$ has exactly two non-degenerate isolated zeros,*

$$P_{\pm} = (\pm t_0, u_0), \quad (17)$$

coinciding with the lateral singularities of Theorem 2.3. Their Jacobian determinants satisfy

$$\det(DF)|_{P_{\pm}} = \mp \frac{5s_0(1-c_0)^2}{Q_0}, \quad (18)$$

where $s_0 = \sin t_0 > 0$, $Q_0 = Q(t_0) > 0$. Hence $\text{Ind}(P_-) = +1$, $\text{Ind}(P_+) = -1$, total charge = 0: $\mathcal{S}$ carries a topological vortex–antivortex pair.

Proof. Zeros. $F = 0$ requires $N = M = 0$. For $c \neq 1$: $N = 0$ gives $u = (1-c)/2$; substituting into $M = 0$ and cancelling $(1+c)/2 > 0$ yields $c^2(2-c) = (1-c)^3$, i.e. $c^2 - 3c + 1 = 0$. The unique root $c_0 = (3 - \sqrt{5})/2$ gives $u_0 = (\sqrt{5} - 1)/4$, reproducing the lateral singularities.

Jacobian at P_+ . Using $c_0^2 = 3c_0 - 1$ throughout:

$$\begin{aligned} N_u &= -2(1-c_0), \\ N_t &= 2s_0(1-c_0-u_0) = s_0(1-c_0), \quad (u_0 = (1-c_0)/2), \\ M_u &= -(1-c_0+c_0^2)/Q_0 = -2c_0/Q_0, \quad (1-c_0+c_0^2 = 2c_0). \end{aligned}$$

Writing $M = P/Q$ with $P = (1-u)c^2(2-c) - u(1-c)^2(1+c)$ and noting $P(c_0, u_0) = 0$:

$$M_t|_{P_+} = \frac{P_t}{Q_0} = \frac{-s_0 \partial_c P}{Q_0}.$$

Evaluating $\partial_c P = (1-u)(4c - 3c^2) + u(1-c)(1+3c)$ at (c_0, u_0) with $c_0^2 = 3c_0 - 1$:

$$4c_0 - 3c_0^2 = 3 - 5c_0, \quad (1-c_0)(1+3c_0) = 4 - 7c_0,$$

so $\partial_c P|_{P_+} = \frac{1+c_0}{2}(3-5c_0) + \frac{1-c_0}{2}(4-7c_0) = (5-7c_0)/2$, giving $M_t|_{P_+} = -s_0(5-7c_0)/(2Q_0)$.

Assembling:

$$\begin{aligned} \det(DF)|_{P_+} &= N_t M_u - N_u M_t \\ &= s_0(1-c_0) \cdot \frac{-2c_0}{Q_0} - (-2(1-c_0)) \cdot \frac{-s_0(5-7c_0)}{2Q_0} \\ &= \frac{-s_0(1-c_0)}{Q_0} [2c_0 + (5-7c_0)] = -\frac{5s_0(1-c_0)^2}{Q_0} < 0. \end{aligned}$$

Parity at P_- . Since N, M depend on t only through c (even) and Q (even), N_t, M_t are odd in t while N_u, M_u are even. Hence $\det(DF)|_{P_-} = -\det(DF)|_{P_+} > 0$, giving $\text{Ind}(P_-) = +1$. $\square$

Figure 4 — Topological Dipole: Phase and Modulus

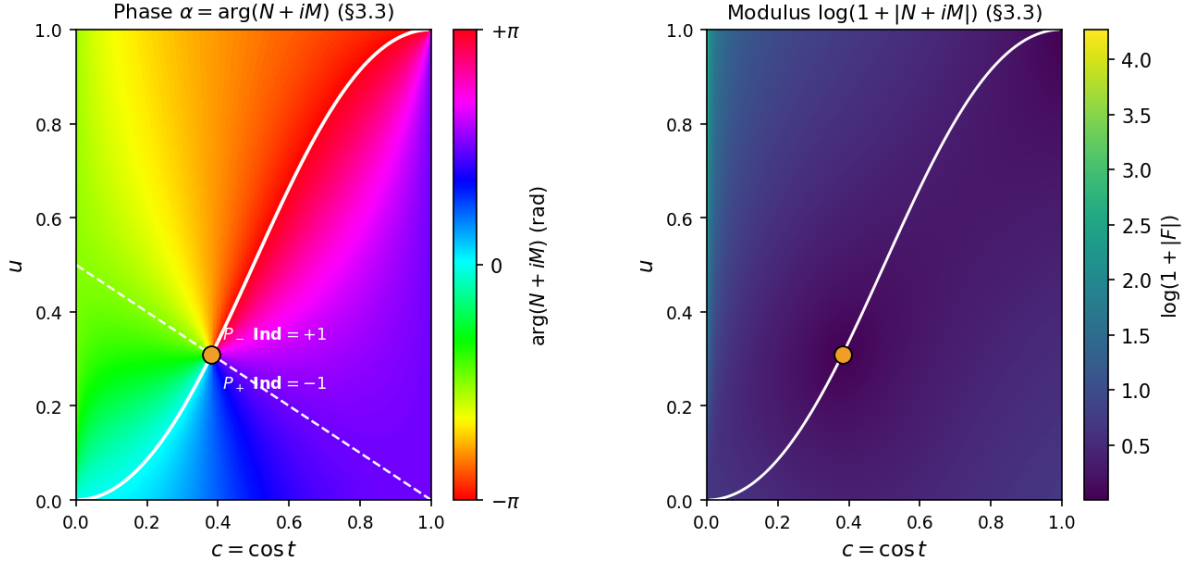


Figure 6: Left: phase $\alpha = \arg(N + iM)$ (HSV colourmap). Winding sense reverses at $P_{\pm}$ (orange dots), confirming opposite indices. Solid white: phase skeleton $M = 0$; dashed white: $N = 0$. Right: modulus $\log(1 + |F|)$; zeros appear as dark wells.

4.4 Triple-Angle Fold Locus

Theorem 4.3 (Triple-angle fold locus). *The fold degeneracy locus $\mathcal{L}_{\text{fold}} := \{J_F = N_t M_u - N_u M_t = 0\}$ of the observation map $(t, u) \mapsto (N, M)$ is parametrized by*

$$u_{\text{fold}}(c) = -\frac{(1-c)c(2-c)(1+2c-c^2)}{c^3-3c+1}. \quad (19)$$

The denominator satisfies the triple-angle identity

$$c^3 - 3c + 1 = 2 \cos(3\theta) + 1 \quad (c = 2 \cos \theta), \quad (20)$$

with unique root $c_ = 2 \cos(4\pi/9) \approx 0.347$ in $(0, 1)$. This root forces a vertical asymptote, bifurcating $\mathcal{L}_{\text{fold}}$ into two disconnected branches; only the branch $c > c_*$ enters the physical domain $u \in [0, 1]$.*

Proof. Step 1: derivatives of N . From $N = (1-c)(1-c-2u)$: $N_u = -2(1-c)$ and $N_t = 2s(1-c-u)$.

Step 2: M_u . Differentiating $M = P/Q$ with $P = (1-u)c^2(2-c) - u(1-c)^2(1+c)$:

$$M_u = \frac{-c^2(2-c) - (1-c)^2(1+c)}{Q} = -\frac{1-c+c^2}{Q},$$

using the denominator identity from Proposition 4.1.

Step 3: M_t . By the quotient rule and (3): $M_t = P_t/Q + sP(1-c)/Q^3$, with $P_t = -s \partial_c P$ and $\partial_c P = (1-u)(4c-3c^2) + u(1-c)(1+3c)$.

Step 4: assembling J_F . After collecting the terms of $J_F = N_t M_u - N_u M_t$ and multiplying by $-Q^3/(2s)$, the expression separates as $A(c) + u B(c)$, where the constant term (set $u = 0$) factors as

$$A(c) = (1-c)c(2-c) \left[(1-c+c^2) + (4c-3c^2) - c(1-c) \right] = (1-c)c(2-c)(1+2c-c^2),$$

and the coefficient of u evaluates to

$$B(c) = c^2(2-c)^2 - (1-c)^2(1-c+c^2) = (4c^2-4c^3+c^4) - (1-3c+4c^2-3c^3+c^4) = c^3-3c+1.$$

Hence $J_F = -\frac{2s}{Q^3}[A(c) + u B(c)]$, and $J_F = 0$ (with $s, Q \neq 0$) gives (19). The triple-angle identity (20) follows from $4 \cos^3 \theta - 3 \cos \theta = \cos(3\theta)$ with $c = 2 \cos \theta$. $\square$

Figure 3 — Parameter Domain: Phase Skeleton, Fold Locus, and Sign of J_F

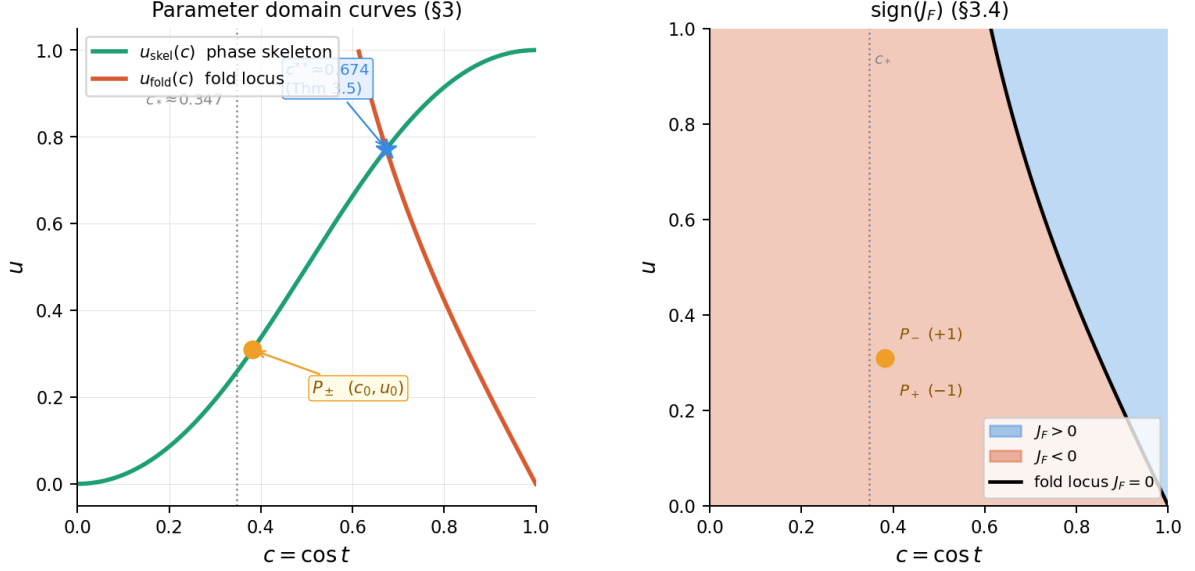


Figure 7: Left: phase skeleton u_{skel} (green) and fold locus u_{fold} (orange) in the (c, u) -plane; $\star$ marks the absolute singularity (Theorem 4.4) at $c^{**} \approx 0.674$. Right: sign of J_F (blue > 0 , salmon < 0); the fold locus (black boundary) separates the two sheets; $P_{\pm}$ lie at $c_0 \approx 0.382$ with indices ± 1 . Dotted vertical: asymptote $c_* \approx 0.347$.

4.5 Absolute Singularity: Skeleton–Fold Intersection

Theorem 4.4 (Absolute singularity). *The intersection condition $u_{\text{skel}}(c) = u_{\text{fold}}(c)$ is equivalent to the quintic equation*

$$c^5 - 3c^4 + 5c^3 - 6c^2 + c + 1 = 0. \quad (21)$$

Since $S(0) = 1 > 0$ and $S(1) = -1 < 0$, the Intermediate Value Theorem guarantees at least one root $c^{**} \in (0, 1)$ ($c^{**} \approx 0.67$), confirming an absolute singularity where observation degeneracy and phase-flow separation coincide.

Proof. Equate (16) and (19); cancel $c(2-c) > 0$: $c/(1-c+c^2) = -(1-c)(1+2c-c^2)/(c^3-3c+1)$. Cross-multiplying and expanding the right side $(1+2c-c^2)(1-c+c^2)(1-c) = 1-3c^2+5c^3-4c^4+c^5$ then moving all terms to one side yields (21). $\square$

5 Signal Geometry and Pseudo-Recurrence

5.1 Whitney Fold Normal Form

Lemma 5.1 (Local double cover). *At each smooth point of $\mathcal{L}_{\text{fold}}$, the observation map $F = (N, M) : (t, u) \rightarrow \mathbb{R}^2$ is locally diffeomorphic to the Whitney fold normal form [2, 3]*

$$(\xi, \eta) \mapsto (\xi, \eta^2). \quad (22)$$

Consequently: (i) the compression rate $1/|J_F|$ diverges to infinity at $\mathcal{L}_{\text{fold}}$; and (ii) the map loses local injectivity—both $(\xi_0, +\delta)$ and $(\xi_0, -\delta)$ map to (ξ_0, δ^2) for any $\delta \neq 0$.

Proof. We verify Whitney’s two conditions.

Smooth fold locus. From the proof of Theorem 4.3: on $\mathcal{L}_{\text{fold}}$ (where $B(c) \neq 0$ for $c \neq c_*$), $\partial_u(A + uB) = B \neq 0$, so $\nabla J_F \neq 0$ and $\mathcal{L}_{\text{fold}}$ is smooth.

Non-degenerate fold. The kernel of DF on $\mathcal{L}_{\text{fold}}$ is spanned by $v = (N_u, -N_t)^\top$. Computing $dJ_F(v)$ via $J_F \propto A + uB$ gives a non-zero result for $s \neq 0$ and $B \neq 0$, confirming v is transversal to $\mathcal{L}_{\text{fold}}$. Whitney’s theorem then provides coordinates (22); (i) and (ii) are immediate consequences. $\square$

Figure 5 — Whitney Fold Normal Form $(\xi, \eta) \mapsto (\xi, \eta^2)$ (Lemma 4.1)

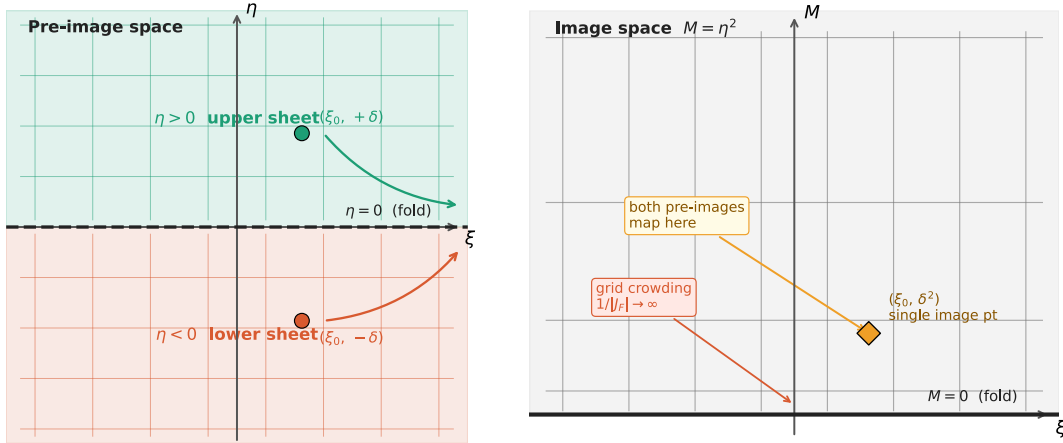


Figure 8: Whitney fold normal form (22). Left: two pre-image sheets (teal $\eta > 0$, coral $\eta < 0$); the symmetric pair $(\xi_0, \pm\delta)$ yields one image. Right: image space; grid crowding near $\eta^2 = 0$ visualises the diverging compression rate $1/|J_F|$.

5.2 Geometry-Induced Frequency Doubling

Proposition 5.2 (Frequency doubling). *In Whitney coordinates near $\mathcal{L}_{\text{fold}}$, suppose the transversal coordinate oscillates as $\eta(t) = a \cos(\omega t)$ with $\xi = \xi_0$ fixed. The observed phase $\phi_{\text{obs}} = \arctan(\eta^2/\xi)$ satisfies*

$$\phi_{\text{obs}}(t) = \arctan(\kappa \cos^2(\omega t)) \approx \frac{\kappa}{2} (1 + \cos(2\omega t)), \quad \kappa := a^2/\xi_0 \ll 1. \quad (23)$$

The dominant observed frequency is 2ω , not ω . This doubling is a pure geometric artifact of the $\mathbb{Z}_2$ fold symmetry $\eta \mapsto -\eta$, independent of any physical oscillator.

Proof. Substitute $\cos^2(\omega t) = (1 + \cos(2\omega t))/2$ and linearize $\arctan(x) \approx x$ for $|x| = \kappa \cos^2(\omega t) \ll 1$. The absence of an ω -component follows from $F(\xi, \eta) = F(\xi, -\eta)$, which suppresses all odd harmonics. $\square$

5.3 Apparent Periodicity

Theorem 5.3 (Pseudo-recurrence and apparent periodicity). *Let $\gamma(s) = (t(s), u(s))$ be a non-periodic intrinsic trajectory crossing $\mathcal{L}_{\text{fold}}$ transversally, and let*

$$\Psi_{\text{obs}}(s; \theta) = A_0(t, u) e^{i(\theta + \phi_0(t, u))}, \quad \theta = \omega_0 s,$$

be the projected signal under uniform rotation. Then:

1. (Pseudo-recurrence) $\exists s_1 \neq s_2$ such that $\Psi_{\text{obs}}(s_1) = \Psi_{\text{obs}}(s_2)$.
2. (Frequency doubling) The dominant Fourier frequency of Ψ_{obs} is 2ω , not any intrinsic frequency of γ (Proposition 5.2).
3. (Geometric origin) Both phenomena arise solely from the Whitney fold geometry; no differential equation governing γ is required.

Proof. (1) As γ crosses $\mathcal{L}_{\text{fold}}$, there exists s_0 with $\eta(s_0) = 0$. By continuity, choose s_1, s_2 such that $\eta(s_1) = +\delta$ and $\eta(s_2) = -\delta$ for the same small $\delta > 0$ and the same ξ -value. Lemma 5.1(ii) gives $F(\xi, +\delta) = F(\xi, -\delta)$, hence $\Psi_{\text{obs}}(s_1) = \Psi_{\text{obs}}(s_2)$.

(2) Immediate from Proposition 5.2.

(3) $F = (N, M)$ with $N = V_z$, $M = V_x$ is a purely geometric object; its fold locus is determined by the surface geometry alone. $\square$

Figure 6 — Geometry-Induced Frequency Doubling and Pseudo-Recurrence (§4.3–4.4)

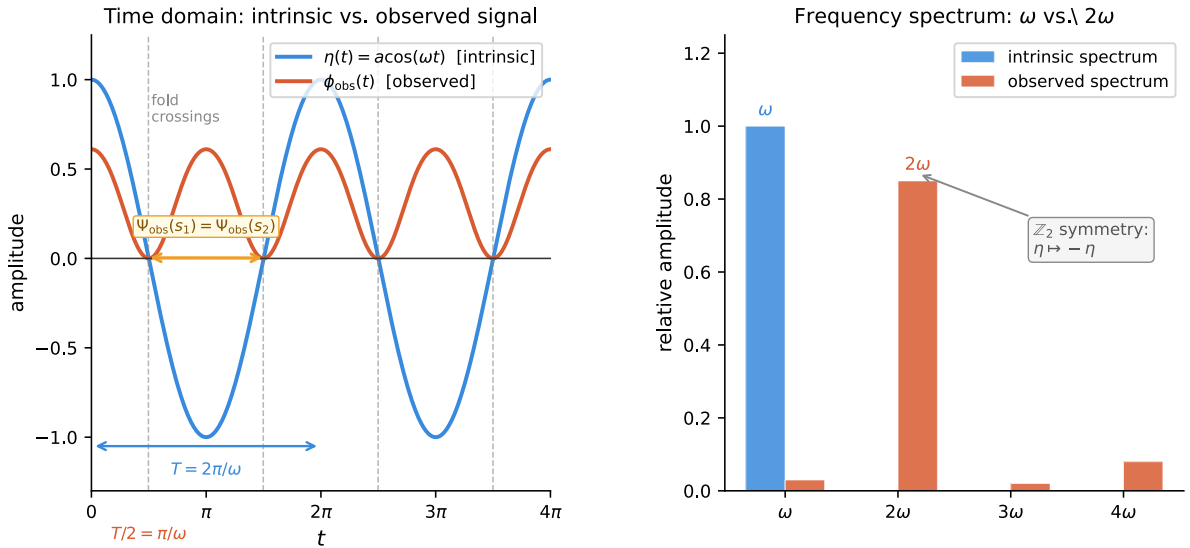


Figure 9: Left: intrinsic signal $\eta(t) = a \cos(\omega t)$ (blue) and observed phase ϕ_{obs} (coral) at half the period; vertical dashes mark fold crossings; the bracket shows a pseudo-recurrence pair (s_1, s_2) from Theorem 5.3. Right: frequency spectrum—intrinsic peak at ω , dominant observed peak at 2ω ($\mathbb{Z}_2$ label).

6 Conclusion

Table 1 summarises all main results and the polynomial controlling each.

Table 1: Summary of principal results and their controlling polynomials.

Result	Ref.	Key formula	Polynomial
Non-developability	Thm 2.1	$\det(S_t, S_u, S_{tu}) = -(c^2 - 3c + 1)s/Q$	$c^2 - 3c + 1$
First fundamental form	Prop 2.2	$G = 2c^2 - 6c + 5 \in [1, 5]$	—
Regularity classification	Thm 2.3	$V = 0$ at 3 isolated points	$c^2 - 3c + 1$
Phase skeleton	Prop 4.1	$u_{\text{skel}} = c^2(2-c)/(1-c+c^2)$	$1 - c + c^2$
Topological dipole	Thm 4.2	$\text{Ind}(P_{\pm}) = \mp 1, \det(DF) = \mp 5s_0(1-c_0)^2/Q_0$	$c^2 - 3c + 1$
Fold locus	Thm 4.3	u_{fold} via $c^3 - 3c + 1$	$c^3 - 3c + 1$
Absolute singularity	Thm 4.4	$c^5 - 3c^4 + 5c^3 - 6c^2 + c + 1 = 0$	quintic
Whitney fold	Lem 5.1	$(\xi, \eta) \mapsto (\xi, \eta^2)$	—
Frequency doubling	Prop 5.2	$\omega \mapsto 2\omega$	$\mathbb{Z}_2$
Pseudo-recurrence	Thm 5.3	$\Psi_{\text{obs}}(s_1) = \Psi_{\text{obs}}(s_2)$	fold geometry

The central finding is that $\mathcal{P}(c) = c^2 - 3c + 1$ is the *core algebraic invariant* of $\mathcal{S}$: its root $c_0 = \varphi^{-2}$ simultaneously marks the boundary of non-developability, the immersion singular locus, and the zeros of the complex observation field. This triple coincidence, arising from three independent derivations, is not a computational accident but reflects a deep structural symmetry of the surface. A complementary polynomial $c^3 - 3c + 1$ —identifiable via the triple-angle identity—separately governs the fold locus, and their interaction produces the quintic absolute singularity.

The signal-geometric results show that the Whitney fold is sufficient to produce frequency doubling and apparent periodicity from purely geometric rigid-body motion, without invoking any differential equation or physical oscillator.

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